

http						
No.	Time	Source	Destination	Protocol	Length	Info
363	4.549343	192.168.10.73	103.230.106.197	HTTP	454	GET / HTTP/1.1
374	4.576510	103.230.106.197	192.168.10.73	HTTP	1494	Continuation
375	4.576512	103.230.106.197	192.168.10.73	HTTP	1208	[TCP Previous segment not captured] Continuation
376	4.576513	103.230.106.197	192.168.10.73	HTTP	1494	[TCP Retransmission] Continuation
382	4.576522	103.230.106.197	192.168.10.73	HTTP	1453	Continuation
397	4.592244	192.168.10.73	103.230.106.197	HTTP	512	GET /css/style.css HTTP/1.1
399	4.595027	192.168.10.73	103.230.106.197	HTTP	624	GET /images/logo_dpe.jpg HTTP/1.1
402	4.598706	103.230.106.197	192.168.10.73	HTTP	383	HTTP/1.1 304 Not Modified
405	4.600687	103.230.106.197	192.168.10.73	HTTP	383	HTTP/1.1 304 Not Modified
409	4.602363	192.168.10.73	103.230.106.197	HTTP	630	GET /images/teletalk_logo.png HTTP/1.1
413	4.608710	103.230.106.197	192.168.10.73	HTTP	384	HTTP/1.1 304 Not Modified

1.

- Frame 363: Packet, 454 bytes on wire (3632 bits), 454 bytes captured (3632 bits) on interface en0, id 0 - Section number: 1 - Interface id: 0 (en0) - Interface name: en0 - Interface description: Wi-Fi - Encapsulation type: Ethernet (1) - Arrival Time: Jul 20, 2026 23:32:10.362166000 +06 - UTC Arrival Time: Jul 20, 2026 17:32:10.362166000 UTC - Epoch Arrival Time: 1784568730.362166000 [Time shift for this packet: 0.000000000 seconds] [Time delta from previous captured frame: 640.000 microseconds] [Time since reference or first frame: 4.549343000 seconds] - Frame Number: 363 - Frame Length: 454 bytes (3632 bits) - Capture Length: 454 bytes (3632 bits) [Frame is marked: False] [Frame is ignored: False] [Protocols in frame: eth:ethertype:ip:tcp:http] - Character encoding: ASCII (0) [Coloring Rule Name: HTTP] [Coloring Rule String: http tcp.port == 80 http2]
- Internet Protocol Version 4, Src: 192.168.10.73, Dst: 103.230.106.197 0100 = Version: 4 0101 = Header Length: 20 bytes (5) - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT) 0000 00.. = Differentiated Services Codepoint: Default (0) 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0) - Total Length: 440 - Identification: 0x0000 (0) 010. = Flags: 0x2, Don't fragment 0... = Reserved bit: Not set .1.. = Don't fragment: Set ..0. = More fragments: Not set ...0 0000 0000 0000 = Fragment Offset: 0 - Time to Live: 64 - Protocol: TCP (6) - Header Checksum: 0x9ba3 [validation disabled] [Header checksum status: Unverified] - Source Address: 192.168.10.73 - Destination Address: 103.230.106.197 [Stream index: 16]

Transmission Control Protocol, Src Port: 51917, Dst Port: 80, Seq: 1, Ack: 1, Len: 388

Source Port: 51917
Destination Port: 80
[Stream index: 8]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (12)]
 ...0. = RST: Absent
 ...0. = FIN: Absent
 1... = Data: Present
 1.. = ACK: Present
 0. = SYN-ACK: Absent
 0 = SYN: Absent
 [Completeness Flags: ..DA..]
[TCP Segment Len: 388]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4162131790
[Next Sequence Number: 389 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3959604446
1000 = Header Length: 32 bytes (8)
[Flags: 0x018 (PSH, ACK)]
 000. = Reserved: Not set
 ...0 = Accurate ECN: Not set
 0... = Congestion Window Reduced: Not set
 0.. = ECN-Echo: Not set
 0. = Urgent: Not set
 1 = Acknowledgment: Set
 1... = Push: Set
 0.. = Reset: Not set
 0. = Syn: Not set
 0 = Fin: Not set
 [TCP Flags:AP...]
Window: 65535
[Calculated window size: 65535]
[Window size scaling factor: -1 (unknown)]
Checksum: 0x6145 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
 > TCP Option - No-Operation (NOP)
 > TCP Option - No-Operation (NOP)
 > TCP Option - Timestamps: TSval 990632494, TSecr 2157278087
[Timestamps]
 [Time since first frame in this TCP stream: 0.00000000 seconds]
 [Time since previous frame in this TCP stream: 0.00000000 seconds]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (388 bytes)

Hypertext Transfer Protocol

GET / HTTP/1.1\r\n
Request Method: GET
Request URI: /
Request Version: HTTP/1.1
Host: dpe.teletalk.com.bd\r\n
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/26.5.2 Safari/605.1.15\r\n
Upgrade-Insecure-Requests: 1\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
Accept-Language: en-US,en;q=0.9\r\n
Priority: u=0, i\r\n
Accept-Encoding: gzip, deflate\r\n
Connection: keep-alive\r\n
\r\n
[Full request URI: http://dpe.teletalk.com.bd/]

Ethernet II, Src: 12:5c:e5:c2:4b:26 (12:5c:e5:c2:4b:26), Dst: CudyTechnolo_03:06:aa (80:af:ca:03:06:aa)

Destination: CudyTechnolo_03:06:aa (80:af:ca:03:06:aa)
 0. = LG bit: Globally unique address (factory default)
 0 = IG bit: Individual address (unicast)
Source: 12:5c:e5:c2:4b:26 (12:5c:e5:c2:4b:26)
 1. = LG bit: Locally administered address (this is NOT the factory default)
 0 = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]

2.

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  Ethernet II, Src: CudyTechnolo_03:06:aa (80:af:ca:03:06:aa), Dst: 12:5c:e5:c2:4b:26 (12:5c:e5:c2:4b:26)
  Destination: 12:5c:e5:c2:4b:26 (12:5c:e5:c2:4b:26)
    ....01. .... = LG bit: Locally administered address (this is NOT the factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: CudyTechnolo_03:06:aa (80:af:ca:03:06:aa)
    ....0. .... = LG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]

  Frame 374: Packet, 1494 bytes on wire (11952 bits), 1494 bytes captured (11952 bits) on interface en0, id 0
    Section number: 1
    Interface id: 0 (en0)
      Interface name: en0
      Interface description: Wi-Fi
      Encapsulation type: Ethernet (1)
      Arrival Time: Jul 20, 2026 23:32:10.389333000 +06
      UTC Arrival Time: Jul 20, 2026 17:32:10.389333000 UTC
      Epoch Arrival Time: 1784568730.389333000
      [Time shift for this packet: 0.000000000 seconds]
      [Time delta from previous captured frame: 1.000 microseconds]
      [Time delta from previous displayed frame: 27.167000 milliseconds]
      [Time since reference or first frame: 4.576510000 seconds]
      Frame Number: 374
      Frame Length: 1494 bytes (11952 bits)
      Capture Length: 1494 bytes (11952 bits)
      [Frame is marked: False]
      [Frame is ignored: False]
      [Protocols in frame: eth:ethertype:ip:tcp:http:data]
      Character encoding: ASCII (0)
      [Coloring Rule Name: HTTP]
      [Coloring Rule String: http || tcp.port == 80 || http2]

  Internet Protocol Version 4, Src: 103.230.106.197, Dst: 192.168.10.73
    0100 .... = Version: 4
    ....0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    ....000 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 1480
  Identification: 0x873e (34622)
  010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 246
  Protocol: TCP (6)
  Header Checksum: 0x5a54 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 103.230.106.197
  Destination Address: 192.168.10.73
  [Stream index: 16]
```


- **Application Layer (HTTP):** The client's web browser uses the GET / HTTP/1.1 request method to ask the server for the root web page document. The application header specifies the destination domain Host: dpe.teletalk.com.bd along with user-agent details (Mozilla/5.0).
- **Transport Layer (TCP):** The transport layer provides process-to-process delivery. It encapsulates the HTTP request into a TCP segment using dynamic Source Port 51917 (identifying the client browser) and well-known Destination Port 80 (identifying the HTTP service on the web server). It sets the PSH and ACK flags to instruct the receiving server to immediately push the data to the server application.
- **Network Layer (IPv4):** The network layer handles end-to-end routing. It encapsulates the TCP segment into an IP packet, attaching Source IP 192.168.10.73 (client) and Destination IP 103.230.106.197 (web server). It specifies protocol type TCP (6) and sets a Time to Live (TTL) of 64.
- **Data Link Layer (Ethernet II):** The data link layer handles local node-to-node frame delivery. It encapsulates the IP packet into an Ethernet frame, adding the client's physical address as the Source MAC (12:5c:e5:c2:4b:26) and the gateway/router's address as the Destination MAC (80:af:ca:03:06:aa).
- **Physical Layer (Frame):** The physical layer converts the 454-byte (3632-bit) frame into actual wireless/electrical signals to be transmitted across the network interface (en0 / Wi-Fi).

2. Operation of Layers for the HTTP Response (Frame 374)

- **Physical Layer (Frame):** The physical layer receives the incoming raw signal on interface en0 (Wi-Fi) and converts it into a digital frame of 1494 bytes (11952 bits).
- **Data Link Layer (Ethernet II):** The data link layer checks local hardware addressing. The frame has Source MAC 80:af:ca:03:06:aa (gateway/router) and Destination MAC 12:5c:e5:c2:4b:26 (client NIC). Upon verifying that the destination MAC matches the local device, it strips the Ethernet header and passes the payload up.
- **Network Layer (IPv4):** The network layer checks logical addressing. The packet header indicates Source IP 103.230.106.197 (web server) and Destination IP 192.168.10.73 (client host). Confirming that the IP matches the local host, it strips the IP header and forwards the TCP segment to the transport layer.
- **Transport Layer (TCP):** The transport layer handles process demultiplexing. It reads Source Port 80 (HTTP server) and Destination Port 51917 to route the incoming segment to the exact browser tab that initiated the connection. It processes the sequence and acknowledgement numbers (Seq: 5793, Ack: 389) to guarantee ordered, reliable data delivery.
- **Application Layer (HTTP):** The transport layer reassembles the TCP payload (1448 bytes of web file data) and presents it to the application layer. The browser parses this returned data to render the requested web page content to the user.